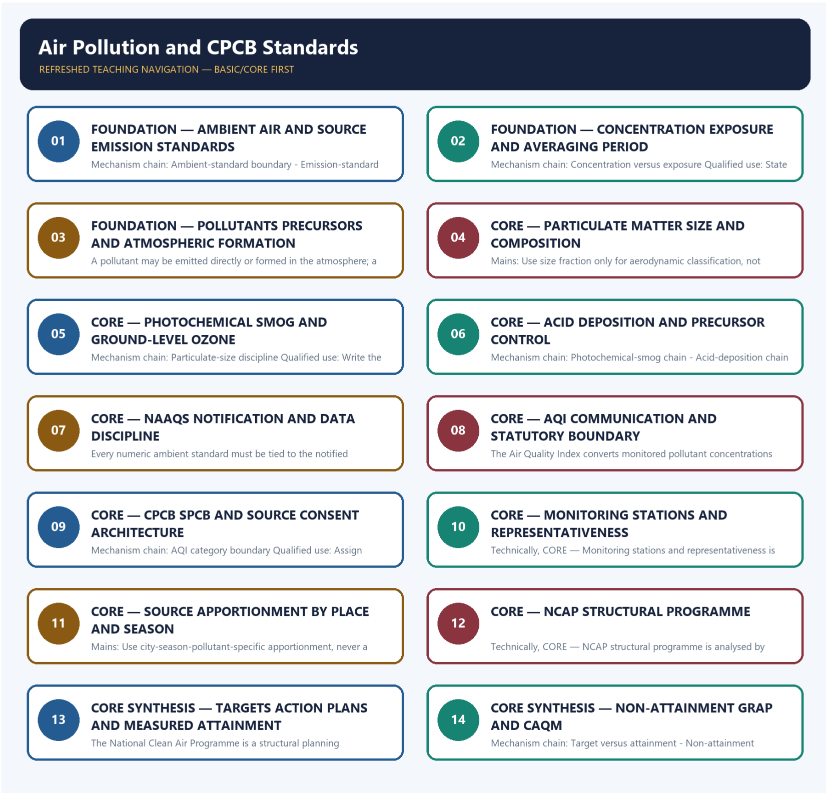
SOLVED PRACTICE WORKBOOK

Air Pollution and CPCB Standards - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



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Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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Air Pollution and CPCB Standards - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Ambient-standard boundary?

A. A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process. B. Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact. C. A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category. D. A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself.

Answer: A. Explanation: A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q2. Which option preserves the ecological boundary of Ambient-standard boundary?

A. Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact. B. A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process. C. Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control. D. A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself.

Answer: B. Explanation: A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q3. Which statement uses Ambient-standard boundary without changing its scale, parameter or status?

A. PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone. B. A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself. C. A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process. D. Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control.

Answer: C. Explanation: A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q4. Which option avoids the standard UPSC close-option trap about Ambient-standard boundary?

A. PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone. B. Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone. C. Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control. D. A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process.

Answer: D. Explanation: A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q5. Which statement correctly identifies Emission-standard boundary?

A. A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category. B. Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact. C. A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself. D. Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control.

Answer: A. Explanation: A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q6. Which option preserves the ecological boundary of Emission-standard boundary?

A. A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself. B. A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category. C. PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone. D. Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control.

Answer: B. Explanation: A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q7. Which statement uses Emission-standard boundary without changing its scale, parameter or status?

A. Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control. B. Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone. C. A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category. D. PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone.

Answer: C. Explanation: A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q8. Which option avoids the standard UPSC close-option trap about Emission-standard boundary?

A. Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition. B. Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone. C. PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone. D. A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category.

Answer: D. Explanation: A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q9. Which statement correctly identifies Concentration versus exposure?

A. Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact. B. PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone. C. Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control. D. A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself.

Answer: A. Explanation: Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q10. Which option preserves the ecological boundary of Concentration versus exposure?

A. Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone. B. Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact. C. PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone. D. Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control.

Answer: B. Explanation: Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q11. Which statement uses Concentration versus exposure without changing its scale, parameter or status?

A. Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone. B. PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone. C. Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact. D. Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition.

Answer: C. Explanation: Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status

categories.

Q12. Which option avoids the standard UPSC close-option trap about Concentration versus exposure?

A. Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page. B. Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition. C. Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone. D. Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact.

Answer: D. Explanation: Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q13. Which statement correctly identifies Pollutant versus precursor?

A. A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself. B. Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control. C. Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone. D. PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone.

Answer: A. Explanation: A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q14. Which option preserves the ecological boundary of Pollutant versus precursor?

A. Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone. B. A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself. C. PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone. D. Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition.

Answer: B. Explanation: A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q15. Which statement uses Pollutant versus precursor without changing its scale, parameter or status?

A. Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition. B. Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone. C. A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself. D. Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page.

Answer: C. Explanation: A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q16. Which option avoids the standard UPSC close-option trap about Pollutant versus precursor?

A. Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition. B. Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page. C. The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard. D. A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself.

Answer: D. Explanation: A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q17. Which statement correctly identifies Primary and secondary pollution?

A. Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control. B. PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone. C. Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition. D. Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone.

Answer: A. Explanation: Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q18. Which option preserves the ecological boundary of Primary and secondary pollution?

A. Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition. B. Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control. C. Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone. D. Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page.

Answer: B. Explanation: Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q19. Which statement uses Primary and secondary pollution without changing its scale, parameter or status?

A. Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page. B. The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard. C. Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control. D. Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition.

Answer: C. Explanation: Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q20. Which option avoids the standard UPSC close-option trap about Primary and secondary pollution?

A. The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard. B. Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page. C. An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period. D. Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control.

Answer: D. Explanation: Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q21. Which statement correctly identifies Particulate-size discipline?

A. PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone. B. Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone. C. Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition. D. Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page.

Answer: A. Explanation: PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q22. Which option preserves the ecological boundary of Particulate-size discipline?

A. The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard. B. PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone. C. Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page. D. Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition.

Answer: B. Explanation: PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q23. Which statement uses Particulate-size discipline without changing its scale, parameter or status?

A. Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page. B. The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard. C. PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone. D. An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period.

Answer: C. Explanation: PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q24. Which option avoids the standard UPSC close-option trap about Particulate-size discipline?

A. CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions. B. An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period. C. The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard. D. PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone.

Answer: D. Explanation: PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q25. Which statement correctly identifies Photochemical-smog chain?

A. Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone. B. Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page. C. The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard. D. Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition.

Answer: A. Explanation: Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q26. Which option preserves the ecological boundary of Photochemical-smog chain?

A. The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard. B. Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone. C. An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period. D. Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page.

Answer: B. Explanation: Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q27. Which statement uses Photochemical-smog chain without changing its scale, parameter or status?

A. An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period. B. CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions. C. Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone. D. The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard.

Answer: C. Explanation: Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q28. Which option avoids the standard UPSC close-option trap about Photochemical-smog chain?

A. CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions. B. An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period. C. The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument. D. Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone.

Answer: D. Explanation: Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q29. Which statement correctly identifies Acid-deposition chain?

A. Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition. B. Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page. C. The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard. D. An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period.

Answer: A. Explanation: Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q30. Which option preserves the ecological boundary of Acid-deposition chain?

A. An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period. B. Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition. C. CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions. D. The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard.

Answer: B. Explanation: Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q31. Which statement uses Acid-deposition chain without changing its scale, parameter or status?

A. The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument. B. An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period. C. Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition. D. CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions.

Answer: C. Explanation: Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q32. Which option avoids the standard UPSC close-option trap about Acid-deposition chain?

A. CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions. B. The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument. C. A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region. D. Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition.

Answer: D. Explanation: Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q33. Which statement correctly identifies NAAQS data discipline?

A. Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page. B. CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions. C. An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period. D. The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard.

Answer: A. Explanation: Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q34. Which option preserves the ecological boundary of NAAQS data discipline?

A. An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period. B. Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page. C. The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument. D. CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions.

Answer: B. Explanation: Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q35. Which statement uses NAAQS data discipline without changing its scale, parameter or status?

A. The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument. B. A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region. C. Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page. D. CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions.

Answer: C. Explanation: Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q36. Which option avoids the standard UPSC close-option trap about NAAQS data discipline?

A. A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region. B. The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument. C. Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised. D. Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page.

Answer: D. Explanation: Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q37. Which statement correctly identifies AQI communication role?

A. The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard. B. CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions. C. An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period. D. The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument.

Answer: A. Explanation: The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments

or status categories.

Q38. Which option preserves the ecological boundary of AQI communication role?

A. A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region. B. The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard. C. The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument. D. CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions.

Answer: B. Explanation: The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q39. Which statement uses AQI communication role without changing its scale, parameter or status?

A. Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised. B. The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument. C. The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard. D. A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region.

Answer: C. Explanation: The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q40. Which option avoids the standard UPSC close-option trap about AQI communication role?

A. The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent. B. Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised. C. A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region. D. The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard.

Answer: D. Explanation: The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q41. Which statement correctly identifies AQI category boundary?

A. An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period. B. The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument. C. CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions. D. A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region.

Answer: A. Explanation: An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period. The other options

belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q42. Which option preserves the ecological boundary of AQI category boundary?

A. A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region. B. An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period. C. Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised. D. The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument.

Answer: B. Explanation: An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q43. Which statement uses AQI category boundary without changing its scale, parameter or status?

A. A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region. B. Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised. C. An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period. D. The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent.

Answer: C. Explanation: An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q44. Which option avoids the standard UPSC close-option trap about AQI category boundary?

A. The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent. B. An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction. C. Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised. D. An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period.

Answer: D. Explanation: An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q45. Which statement correctly identifies CPCB and SPCB roles?

A. CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions. B. A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region. C. The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument. D. Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised.

Answer: A. Explanation: CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions. The other options belong to different media, parameters, processes,

scales, actors, institutions, instruments or status categories.

Q46. Which option preserves the ecological boundary of CPCB and SPCB roles?

A. The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent. B. CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions. C. A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region. D. Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised.

Answer: B. Explanation: CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q47. Which statement uses CPCB and SPCB roles without changing its scale, parameter or status?

A. Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised. B. The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent. C. CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions. D. An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction.

Answer: C. Explanation: CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q48. Which option avoids the standard UPSC close-option trap about CPCB and SPCB roles?

A. An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction. B. Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI. C. The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent. D. CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions.

Answer: D. Explanation: CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q49. Which statement correctly identifies Air Act and consent layer?

A. The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument. B. Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised. C. The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent. D. A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region.

Answer: A. Explanation: The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal

instrument. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q50. Which option preserves the ecological boundary of Air Act and consent layer?

A. An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction. B. The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument. C. Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised. D. The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent.

Answer: B. Explanation: The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q51. Which statement uses Air Act and consent layer without changing its scale, parameter or status?

A. An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction. B. The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent. C. The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument. D. Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI.

Answer: C. Explanation: The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q52. Which option avoids the standard UPSC close-option trap about Air Act and consent layer?

A. Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI. B. GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India. C. An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction. D. The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument.

Answer: D. Explanation: The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q53. Which statement correctly identifies Monitoring representativeness?

A. A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region. B. Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised. C. The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent. D. An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction.

Answer: A. Explanation: A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q54. Which option preserves the ecological boundary of Monitoring representativeness?

A. Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI. B. A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region. C. An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction. D. The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent.

Answer: B. Explanation: A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q55. Which statement uses Monitoring representativeness without changing its scale, parameter or status?

A. An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction. B. GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India. C. A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region. D. Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI.

Answer: C. Explanation: A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q56. Which option avoids the standard UPSC close-option trap about Monitoring representativeness?

A. GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India. B. Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI. C. Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading. D. A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region.

Answer: D. Explanation: A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q57. Which statement correctly identifies Source-apportionment boundary?

A. Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised. B. Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI. C. An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction. D. The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent.

Answer: A. Explanation: Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q58. Which option preserves the ecological boundary of Source-apportionment boundary?

A. GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India. B. Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised. C. Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI. D. An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction.

Answer: B. Explanation: Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q59. Which statement uses Source-apportionment boundary without changing its scale, parameter or status?

A. Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI. B. Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading. C. Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised. D. GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India.

Answer: C. Explanation: Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q60. Which option avoids the standard UPSC close-option trap about Source-apportionment boundary?

A. Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading. B. A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process. C. GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India. D. Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised.

Answer: D. Explanation: Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q61. Which statement correctly identifies NCAP programme boundary?

A. The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent. B. An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction. C. GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India. D. Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI.

Answer: A. Explanation: The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q62. Which option preserves the ecological boundary of NCAP programme boundary?

A. Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading. B. The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent. C. GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India. D. Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI.

Answer: B. Explanation: The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q63. Which statement uses NCAP programme boundary without changing its scale, parameter or status?

A. GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India. B. Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading. C. The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent. D. A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process.

Answer: C. Explanation: The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q64. Which option avoids the standard UPSC close-option trap about NCAP programme boundary?

A. A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process. B. A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category. C. Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading. D. The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent.

Answer: D. Explanation: The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent. The other options

belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q65. Which statement correctly identifies Target versus attainment?

A. An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction. B. GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India. C. Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading. D. Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI.

Answer: A. Explanation: An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q66. Which option preserves the ecological boundary of Target versus attainment?

A. A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process. B. An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction. C. Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading. D. GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India.

Answer: B. Explanation: An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q67. Which statement uses Target versus attainment without changing its scale, parameter or status?

A. A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process. B. A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category. C. An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction. D. Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading.

Answer: C. Explanation: An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q68. Which option avoids the standard UPSC close-option trap about Target versus attainment?

A. A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category. B. Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact. C. A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process. D. An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction.

Answer: D. Explanation: An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q69. Which statement correctly identifies Non-attainment boundary?

A. Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI. B. GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India. C. Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading. D. A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process.

Answer: A. Explanation: Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q70. Which option preserves the ecological boundary of Non-attainment boundary?

A. A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category. B. Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI. C. A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process. D. Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading.

Answer: B. Explanation: Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q71. Which statement uses Non-attainment boundary without changing its scale, parameter or status?

A. A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category. B. A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process. C. Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI. D. Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact.

Answer: C. Explanation: Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q72. Which option avoids the standard UPSC close-option trap about Non-attainment boundary?

A. A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category. B. A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself. C. Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact. D. Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI.

Answer: D. Explanation: Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q73. Which statement correctly identifies GRAP and CAQM boundary?

A. GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India. B. A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process. C. Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading. D. A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category.

Answer: A. Explanation: GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q74. Which option preserves the ecological boundary of GRAP and CAQM boundary?

A. A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category. B. GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India. C. Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact. D. A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process.

Answer: B. Explanation: GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q75. Which statement uses GRAP and CAQM boundary without changing its scale, parameter or status?

A. Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact. B. A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category. C. GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India. D. A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself.

Answer: C. Explanation: GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or

status categories.

Q76. Which option avoids the standard UPSC close-option trap about GRAP and CAQM boundary?

A. A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself. B. Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact. C. Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control. D. GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India.

Answer: D. Explanation: GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q77. Which statement correctly identifies Audited evidence boundary?

A. Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading. B. A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category. C. A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process. D. Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact.

Answer: A. Explanation: Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q78. Which option preserves the ecological boundary of Audited evidence boundary?

A. A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself. B. Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading. C. Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact. D. A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category.

Answer: B. Explanation: Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q79. Which statement uses Audited evidence boundary without changing its scale, parameter or status?

A. A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself. B. Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact. C. Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading. D. Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control.

Answer: C. Explanation: Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q80. Which option avoids the standard UPSC close-option trap about Audited evidence boundary?

A. Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control. B. A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself. C. PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone. D. Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading.

Answer: D. Explanation: Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

PYQS AND ANSWER PRACTICE

AUDITED AIR-POLLUTION, NCAP, SMOG AND AQI PYQ OWNERSHIP

Audited ledgers route direct Mains demands on NCAP, WHO air-quality guidelines and photochemical smog, plus objective demands on biomass burning, benzene, coal combustion, sulphur and nitrogen precursors, PM-related sources and cloud seeding. Objective keys are not inferred.

OWNER PYQ LEDGER EXTRACTS

9. PYQ application

- ANALYSIS: Recurring Prelims pattern: distinguish the functions of CPCB, SPCBs, CAQM and identify

the correct definition of NAAQS versus AQI.

- ANALYSIS: Mains linkage: the multi-state, multi-sectoral nature of Delhi-NCR's air pollution is

used to argue for coordinated regional governance (illustrated by CAQM's creation).

Recent PYQ Integration (2024-2025)

> **Status:** 2024-2025 question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-PRELIMS-2024-2025.md`. **Answer-key rule:** The official 2024-2025 Prelims Set-A keys are present in the repository and CSAT Set-A keys are supplied; even so, no option or answer is recorded or inferred in this integration.

- **Years represented:** 2024, 2025
- **Paper(s):** Prelims GS-I
- **Routed question demands:** 2

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2024	Prelims GS-I	90	EPA's largest source of sulphur dioxide emissions (fossil-fuel power plants)	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.
2025	Prelims GS-I	85	Artificial rainfall (cloud seeding) to reduce air pollution	Objective question; official Set-A key available locally, answer not inferred	Key available locally (official Set-A answer key present); answer not recorded here	Cover the named fact/concept and its likely statement-level distinctions.

What this owner must now support

- EPA's largest source of sulphur dioxide emissions (fossil-fuel power plants)
- Artificial rainfall (cloud seeding) to reduce air pollution

> This block integrates the 2024-2025 examinable demand and paper metadata. It is kept separate from the 2018-2023 block and does not convert an unkeyed/answer-free objective question into a solved answer.

Historical PYQ Integration (2018-2023)

> **Status:** Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-MAINS-GS3-GS4-2018-2023.md`, `_PYQ-ROUTING-PRELIMS-2018-2023.md`. **Answer-key rule:** The official 2018-2023 Prelims/CSAT keys are not held locally; no option or answer has been inferred.

- **Years represented:** 2019, 2020, 2021, 2022
- **Paper(s):** GS-III, Prelims GS-I
- **Routed question demands:** 11

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2019	Prelims GS-I	35	Pollutants released from burning crop and biomass residue	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2020	GS-III	17	National Clean Air Programme key features and implementation	What are · 15 marks · 250 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2020	Prelims GS-I	48	Benzene pollution exposure sources and contributing factors	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2020	Prelims GS-I	79	Coal ash toxins and coal power plant air emissions	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2021	GS-III	7	WHO revised Air Quality Guidelines and India's NCAP response	Describe · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2021	Prelims GS-I	17	Environmental concerns from copper smelting plants	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2021	Prelims GS-I	18	Furnace oil properties uses and sulphur emissions	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2021	Prelims GS-I	25	Magnetite particles environmental air pollution sources	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.
2022	GS-III	7	Photochemical smog formation effects and Gothenburg Protocol mitigation	Discuss · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2022	Prelims GS-I	44	WHO Air Quality Guidelines PM2.5 and ozone standards	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2022	Prelims GS-I	100	Atmospheric pollutants causing acid rain identification	Objective question; official key unavailable locally	Routed; key unavailable locally	Cover the named fact/concept and its likely statement-level distinctions.

What this owner must now support

- Pollutants released from burning crop and biomass residue
- National Clean Air Programme key features and implementation
- Benzene pollution exposure sources and contributing factors
- Coal ash toxins and coal power plant air emissions
- WHO revised Air Quality Guidelines and India's NCAP response
- Environmental concerns from copper smelting plants
- Furnace oil properties uses and sulphur emissions
- Magnetite particles environmental air pollution sources
- Photochemical smog formation effects and Gothenburg Protocol mitigation
- WHO Air Quality Guidelines PM2.5 and ozone standards
- Atmospheric pollutants causing acid rain identification

> The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

10. PYQ-based analytical application

- ANALYSIS: Prelims questions on CAQM's jurisdiction and statutory basis should be answered using

the specific states covered (Delhi, Punjab, Haryana, UP, Rajasthan) under the 2021 Act.

- ANALYSIS: Mains answers on "air pollution governance" should explicitly distinguish reactive

(GRAP) from structural (NCAP) mechanisms and address the source-apportionment nuance to avoid the common crop-residue-burning overattribution error.

Historical PYQ Integration (2018-2023)

> **Status:** Question-level PYQ demand is integrated into this owner. **Provenance:** Audited local official-paper routing ledgers: `_PYQ-ROUTING-MAINS-GS3-GS4-2018-2023.md`.

- **Years represented:** 2020, 2021, 2022
- **Paper(s):** GS-III
- **Routed question demands:** 3

Year	Paper	Q	PYQ demand (neutral rendering)	Directive / format	Source status	Owner requirement
2020	GS-III	17	National Clean Air Programme key features and implementation	What are · 15 marks · 250 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2021	GS-III	7	WHO revised Air Quality Guidelines and India's NCAP response	Describe · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.
2022	GS-III	7	Photochemical smog formation effects and Gothenburg Protocol mitigation	Discuss · 10 marks · 150 words	Routed to owning topic	Prepare context, core dimensions, evidence/examples, counterpoint and a concise conclusion.

What this owner must now support

- National Clean Air Programme key features and implementation
- WHO revised Air Quality Guidelines and India's NCAP response
- Photochemical smog formation effects and Gothenburg Protocol mitigation

> The table integrates the examinable demand and paper metadata. It does not turn an unkeyed objective question into a solved answer, and it does not claim that lexical presence alone proves full conceptual sufficiency.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish ambient air-quality standards from source-emission standards. Answer in about 150 words.

Model thesis: Claim: Ambient-standard boundary. **Named evidence/example:** A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Emission-standard boundary. **Named evidence/example:** A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NAAQS data discipline. **Named evidence/example:** Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process.
- A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category.
- Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page.

Qualified conclusion: Claim: Ambient-standard boundary. **Named evidence/example:** A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Emission-standard boundary. **Named evidence/example:** A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NAAQS data discipline. **Named evidence/example:** Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

ORIGINAL MAINS 2 - 10 MARKS

Question: Explain why AQI cannot be treated as the NAAQS compliance table. Answer in about 150 words.

Model thesis: Claim: NAAQS data discipline. **Named evidence/example:** Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** AQI communication role. **Named evidence/example:** The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** AQI category boundary. **Named evidence/example:** An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page.
- The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard.
- An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period.

Qualified conclusion: Claim: NAAQS data discipline. **Named evidence/example:** Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** AQI communication role. **Named evidence/example:** The Air Quality Index converts monitored pollutant concentrations into a public communication category; it does not replace the notified ambient standard or a source-emission standard. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** AQI category boundary. **Named evidence/example:** An AQI category describes the index for a stated place and time; it is not by itself a statutory finding that every pollutant met or breached every NAAQS averaging period. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

ORIGINAL MAINS 3 - 15 MARKS

Question: Explain pollutant formation from primary emissions and precursors. Answer in about 250 words.

Model thesis: Claim: Pollutant versus precursor. **Named evidence/example:** A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Primary and secondary pollution. **Named evidence/example:** Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Particulate-size discipline. **Named evidence/example:** PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Photochemical-smog chain. **Named evidence/example:** Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Acid-deposition chain. **Named evidence/example:** Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself.
- Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control.
- PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone.
- Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone.
- Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition.

Qualified conclusion: Claim: Pollutant versus precursor. **Named evidence/example:** A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Primary and secondary pollution. **Named evidence/example:** Primary pollutants are emitted from sources, whereas secondary pollutants form through atmospheric chemistry; the control pathway must therefore include both direct emissions and precursor control. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Particulate-size discipline. **Named evidence/example:** PM2.5 and PM10 are aerodynamic size fractions, not chemical species; composition, source and health effect cannot be inferred from size label alone. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Photochemical-smog chain. **Named evidence/example:** Photochemical smog requires precursor emissions and sunlight-driven chemistry that can form ground-level ozone and other oxidants; ozone at the surface is distinct from protective stratospheric ozone. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Acid-deposition chain. **Named evidence/example:** Sulphur and nitrogen oxides can act as acid-deposition precursors after atmospheric transformation; an answer must separate precursor emission from later wet or dry deposition. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

ORIGINAL MAINS 4 - 15 MARKS

Question: Assess monitoring and source-apportionment limits in city air policy. Answer in about 250 words.

Model thesis: Claim: Concentration versus exposure. **Named evidence/example:** Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Monitoring representativeness. **Named evidence/example:** A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Source-apportionment boundary. **Named evidence/example:** Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact.
- A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region.
- Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised.

Qualified conclusion: Claim: Concentration versus exposure. **Named evidence/example:** Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Monitoring representativeness. **Named evidence/example:** A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Source-apportionment boundary. **Named evidence/example:** Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

ORIGINAL MAINS 5 - 20 MARKS

Question: Evaluate India's statutory, programme and episodic air-governance layers. Answer in about 300 words.

Model thesis: Claim: Ambient-standard boundary. **Named evidence/example:** A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** CPCB and SPCB roles. **Named evidence/example:** CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Air Act and consent layer. **Named evidence/example:** The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NCAP programme boundary. **Named evidence/example:** The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Target versus attainment. **Named evidence/example:** An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Non-attainment boundary. **Named evidence/example:** Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** GRAP and CAQM boundary. **Named evidence/example:** GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process.

- CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions.
- The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument.
- The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent.
- An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction.
- Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI.
- GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India.

Qualified conclusion: Claim: Ambient-standard boundary. **Named evidence/example:** A National Ambient Air Quality Standard applies to pollutant concentration in outdoor receiving air; it is not a source-emission limit for a stack, vehicle or process. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** CPCB and SPCB roles. **Named evidence/example:** CPCB coordinates nationally and supports standards and monitoring frameworks, while SPCBs and pollution-control committees perform major consent, monitoring and enforcement functions within their jurisdictions. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Air Act and consent layer. **Named evidence/example:** The Air Act provides the pollution-control framework, while consent conditions and sector-specific standards regulate sources; a prior environmental clearance is a separate project-appraisal instrument. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NCAP programme boundary. **Named evidence/example:** The National Clean Air Programme is a structural planning programme for sustained source reduction and capacity building, not a substitute for the Air Act or a source consent. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Target versus attainment. **Named evidence/example:** An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Non-attainment boundary. **Named evidence/example:** Non-attainment classification is based on the applicable monitoring and standards framework over a defined period; it is not the same as a city's current hourly or daily AQI. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** GRAP and CAQM boundary. **Named evidence/example:** GRAP is a graded episodic response for the Delhi-NCR problem, while CAQM is the statutory regional coordination institution; neither replaces long-term source reduction across India. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

ORIGINAL MAINS 6 - 20 MARKS

Question: Build a source-to-exposure clean-air strategy without unsupported numbers. Answer in about 300 words.

Model thesis: Claim: Emission-standard boundary. **Named evidence/example:** A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Concentration versus exposure. **Named evidence/example:** Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Pollutant versus precursor. **Named evidence/example:** A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NAAQS data discipline. **Named evidence/example:** Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Monitoring representativeness. **Named evidence/example:** A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Source-apportionment boundary. **Named evidence/example:** Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Target versus attainment. **Named evidence/example:** An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Audited evidence boundary. **Named evidence/example:** Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category.
- Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact.

- A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself.
- Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page.
- A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region.
- Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised.
- An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction.
- Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading.

Qualified conclusion: Claim: Emission-standard boundary. **Named evidence/example:** A source-emission standard controls what a specified source may release under the applicable rule or consent; compliance cannot be inferred from a city AQI category. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Concentration versus exposure. **Named evidence/example:** Ambient concentration is a measured amount in air at a place and averaging period, while exposure also depends on where people are, how long they remain and the route of contact. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Pollutant versus precursor. **Named evidence/example:** A pollutant may be emitted directly or formed in the atmosphere; a precursor participates in reactions that create a secondary pollutant and must not be reported as the resulting pollutant itself. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NAAQS data discipline. **Named evidence/example:** Every numeric ambient standard must be tied to the notified pollutant, unit, averaging period, area applicability and compliance rule; no value is carried here from a title-only live page. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Monitoring representativeness. **Named evidence/example:** A station measures conditions at its location and time; network density, siting, data completeness and meteorology affect whether one reading represents a neighbourhood, city or region. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Source-apportionment boundary. **Named evidence/example:** Source contribution varies by city, season, pollutant, method and meteorology; no fixed vehicle, industry, dust or crop-residue share can be universalised. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Target versus attainment. **Named evidence/example:** An NCAP goal, city action plan, fund release, activity completion and measured attainment are different stages; an announced target is not an achieved concentration reduction. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Audited evidence boundary. **Named evidence/example:** Audited ledgers carry NCAP, WHO-guideline, photochemical-smog, acid-rain, combustion-source and cloud-seeding concepts into practice without inventing an objective key, source share, standard value or current reading. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.